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Bus Architecture And High Speed Interfaces

CSA1220-Computer Architecture

Guided By:
Dr S Radhika

Presented by:
P. Rishitha(192472396)



INTRODUCTION

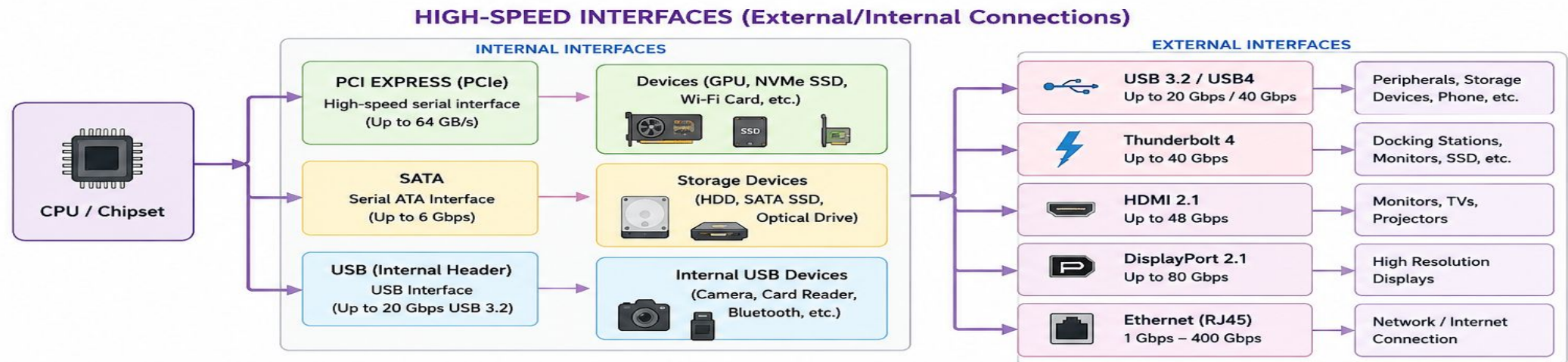
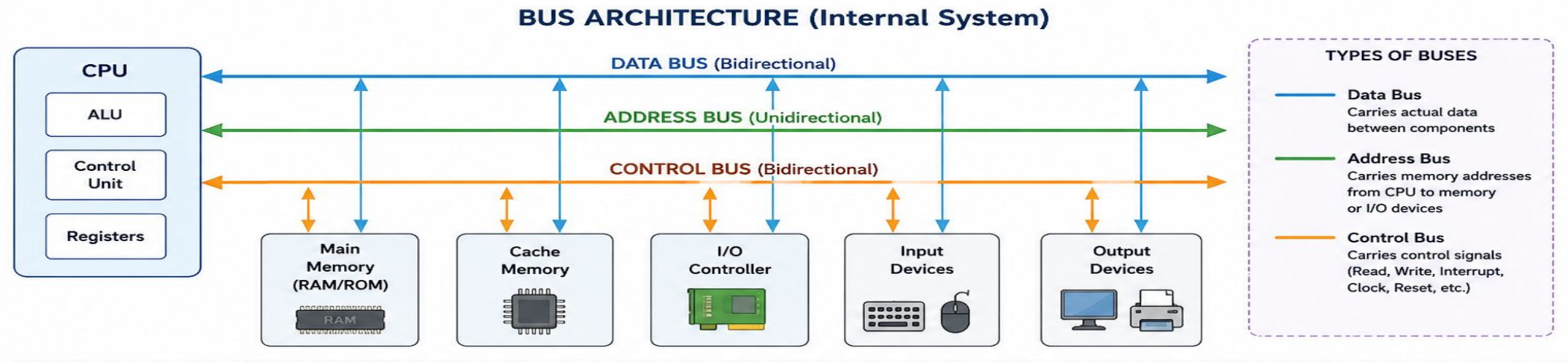
- Bus architecture is a communication system that connects the CPU, memory, and input/output (I/O) devices in a computer.
- It helps transfer data, addresses, and control signals between different hardware components.
- High-speed interfaces are technologies that allow fast data transfer between a computer and external or internal devices.
- Examples of high-speed interfaces include USB, PCI Express (PCIe), SATA, HDMI, and Thunderbolt, which improve the speed and performance of a computer.



Objectives

- To understand the concept of bus architecture.
- To learn the types of buses used in a computer.
- To study high-speed interfaces and their functions.
- To know how data is transferred quickly between computer components and devices.

System Architecture





Components of a Bus

- **Data Bus**

- Transfers data between the CPU, memory, and I/O devices.
- It is **bidirectional** (data moves in both directions).

- **Address Bus**

- Carries the memory address from the CPU to memory or I/O devices.
- It is **unidirectional** (CPU → Memory/I/O).

- **Control Bus**

- Carries control signals such as **Read, Write, Interrupt, and Clock**.
- It controls the operations of the computer.



Types of Bus Architecture

- **Single Bus Architecture:** Uses one bus for communication between all components.
- **Dual Bus Architecture:** Uses two buses to improve data transfer speed.
- **Multiple Bus Architecture:** Uses three or more buses for faster and efficient communication.

High Speed Interfaces

- **USB (Universal Serial Bus):** Connects external devices like keyboards, mice, and pen drives.
- **PCI Express (PCIe):** Connects high-speed internal devices like graphics cards and SSDs.
- **SATA (Serial ATA):** Connects hard disks and SSDs to the motherboard.
- **Thunderbolt:** Provides very fast data, video, and power transfer through a single cable.
- **Ethernet:** Provides high-speed wired network and internet connectivity.



Conclusion

- Bus architecture enables communication between the CPU, memory, and I/O devices.
- High-speed interfaces provide faster and more efficient data transfer.
- They improve the overall performance of a computer system.
- Technologies like USB, PCIe, SATA, HDMI, and Thunderbolt make data communication fast and reliable.

Thank you